

Unit 4

date: 10-10-25
Day: Friday.

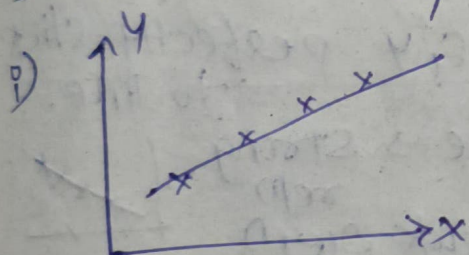
Topics

1. correlation co-efficient
 2. simple linear regression (SLR)
 3. Multiple linear regression (MLR)
 4. Validation techniques for SLR, MLR.
- * It belongs to predictive Data Analytics.

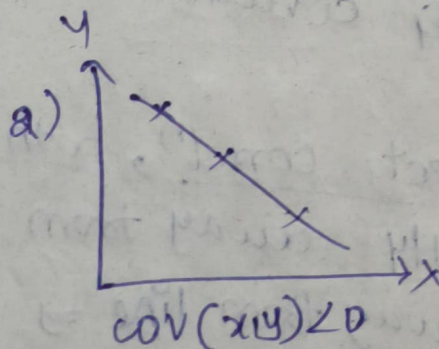
* Covariance

* It measures direction of 2 random variables:

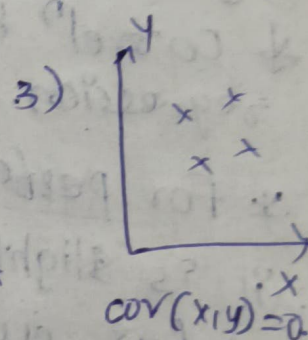
1. positive relationship ($x \uparrow, y \uparrow$ or $x \downarrow, y \downarrow$)
2. Negative relationship ($x \uparrow, y \downarrow$ or $x \downarrow, y \uparrow$)
3. No relationship.



$$\text{cov}(x, y) > 0$$



$$\text{cov}(x, y) < 0$$



$$\text{cov}(x, y) = 0$$

$$\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x}) \cdot (y_i - \bar{y})}{N}$$

= population
(n-1 $\Rightarrow$ sample)

* Range for covariance $\Rightarrow [-\infty \text{ to } +\infty]$

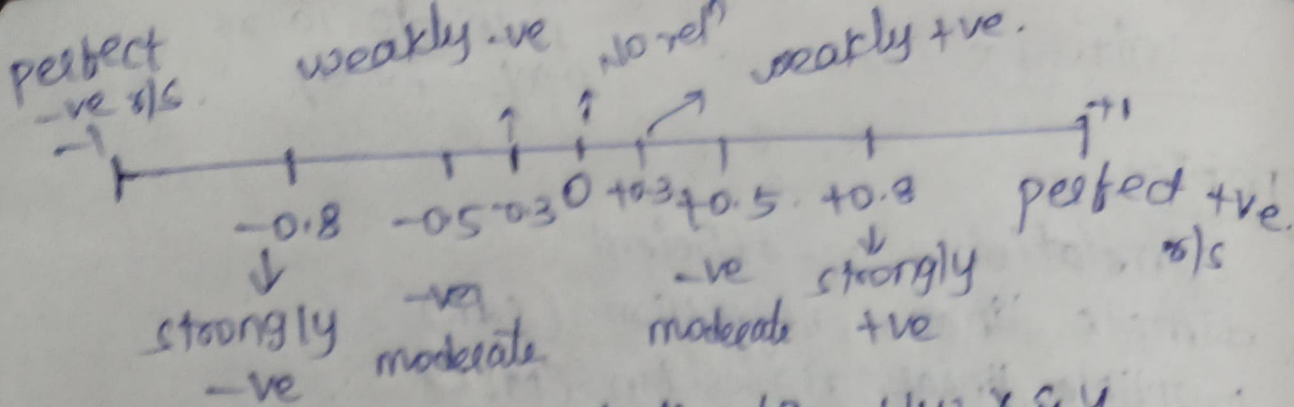
* correlation

* It describes ^{measures} strength & direction of 2 random variables.

* Range for correlation $\Rightarrow [-1 \text{ to } +1]$

* It is also:

1. positive relⁿship
2. Negative relⁿship
3. No relⁿship



$\pm 1 \rightarrow$ perfect $\pm$ relationship b/w x & y

$\pm 0.8 \rightarrow$ strongly $\pm$ relationship

$\pm 0.5 \rightarrow \pm$ moderate

$\pm 0.3 \rightarrow$ weakly $\pm$.

$\pm 0.9 \rightarrow$ very strong $\pm$.

$\rightarrow 0 \rightarrow$ No relⁿ b/w x & y .

* Correlⁿ & covariance are used for linear regression.

* for perfect correlⁿ, both x & y perfectly stick to line.

$\rightarrow$ It is slightly away from line $\rightarrow$ strong relⁿ

$\rightarrow$ far away from line $\rightarrow$ weak relⁿ



* to measure the correlⁿ, we need correlⁿ coefficient else "pearson correlⁿ coefficient (r)".

Formula:

$$\text{Corr}(x, y) \text{ (or) } r(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y}$$

for popⁿ
(σ_x, σ_y for sample)
std devⁿ

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$r(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x}) \cdot (y_i - \bar{y})}{N}$$

$$\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{N}} \cdot \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{N}}$$

$$\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

(OR)

$$\text{cov}(x, y) = \frac{N \cdot \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \cdot \sum_{i=1}^n y_i}{\sqrt{[N \cdot (\sum x^2 - (\sum x)^2)] \cdot [N \cdot (\sum y^2 - (\sum y)^2)]}}$$

$$\sqrt{[N \cdot (\sum x^2 - (\sum x)^2)] \cdot [N \cdot (\sum y^2 - (\sum y)^2)]}$$

① Find the covariance for the following data.

x	y
6	12
5	10
3	8
4	6
2	4

Sol: $\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$

$\bar{x} = 4, \bar{y} = 8$

$\text{cov}(x, y) = \frac{(2 \cdot 4) + (1 \cdot 2) + (-1 \cdot 0) + (0 \cdot -2) + (-2 \cdot -4)}{5}$

$= \frac{8 + 2 + 0 + 0 + 8}{5} = \frac{18}{5} = 3.6$

$\therefore \text{cov}(x, y) = 3.6 > 0$

∴ There is a positive relationship b/w x & y $\left(\begin{matrix} x \uparrow, y \uparrow \\ \text{or} \\ x \downarrow, y \downarrow \end{matrix} \right)$

(Study timing hrs)

x	1	2	3	4
y	10	8	6	4

(No. of mistakes)

Sol: $\bar{x} = 2.5$ & $\bar{y} = 7$

$\text{cov}(x, y) = \frac{(-1.5 \cdot 3) + (-0.5 \cdot 1) + (0.5 \cdot -1) + (1.5 \cdot -3)}{4}$

$\text{cov}(x, y) = \frac{-2.5}{4} = -0.625 \Rightarrow -0.625$

∴ There is a -ve relationship b/w x & y $\left(\begin{matrix} x \uparrow, y \downarrow \\ \text{or} \\ x \downarrow, y \uparrow \end{matrix} \right)$

iii)

X (shoe size)	1	2	3	4
Y (Intelligence)	5	3	7	4

Sol: $\bar{x} = 2.5, \bar{y} = 4.75$

X	Y	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$
1	5	-1.5	0.25	-0.375
2	3	-0.5	-1.75	0.875
3	7	0.5	2.25	1.125
4	4	1.5	-0.75	-1.125

$\text{cov}(x, y) = \frac{-0.375 + 0.875 + 1.125 - 1.125}{4}$

$\boxed{\text{cov}(x, y) = 0.125 > 0} \Rightarrow$ the relⁿ b/w X & Y //

② The std devⁿ of X is 15 & std devⁿ of Y is 20. The correlⁿ b/w X & Y is $\frac{1}{2}$. Find the cov(x, y).

Sol: $\text{corr}(x, y) = \frac{\text{cov}(x, y)}{\sigma_x \cdot \sigma_y}$

$\frac{1}{2} = \frac{\text{cov}(x, y)}{15 \times 20}$

$\text{cov}(x, y) = 15 \times 20 \times \frac{1}{2}$

$\boxed{\text{cov}(x, y) = 1500} //$

③ Find correlⁿ for the following data

X	3	4	6	8	9
Y	4	5	9	10	12

$\bar{x} = 6$

$\bar{y} = 8$

Sol: //

X	Y	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
3	4	-3	-4	12	9	16
4	5	-2	-3	6	4	9
6	9	0	1	0	0	1
8	10	2	2	4	4	4
9	12	3	4	12	9	16

$$\begin{aligned}
 \text{corr}(X, Y) &= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \\
 &= \frac{12 + 6 + 0 + 4 + 12}{\sqrt{9 + 4 + 0 + 4 + 9} \cdot \sqrt{16 + 9 + 1 + 4 + 16}} \\
 &= \frac{34}{\sqrt{26} \cdot \sqrt{46}} = \boxed{0.98} \Rightarrow \text{Very strong} \\
 &\quad \text{+ve relationship} \\
 &\quad \text{B/w } X \text{ \& } Y //
 \end{aligned}$$

4) Find the corrⁿ for the following data

X	5	10	15	20	25
Y	25	24	20	18	15

Sol: $\bar{X} = 15, \bar{Y} = 20.4$

X	Y	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
5	25	-10	4.6	-46	100	21.16
10	24	-5	3.6	-18	25	12.96
15	20	0	-0.4	0	0	0.16
20	18	5	-2.4	-12	25	5.76
25	15	10	-5.4	-54	100	29.16

$$\begin{aligned}
 \text{corr}(X, Y) &= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \\
 &= \frac{-46 - 18 + 0 - 12 - 54}{\sqrt{100 + 25 + 25 + 100} \cdot \sqrt{21.16 + 12.96 + 0.16 + 5.76 + 29.16}} \\
 &= \frac{-130}{\sqrt{250} \sqrt{69.2}} = \boxed{-0.988} \Rightarrow \text{Very strong} \\
 &\quad \text{-ve relationship} \\
 &\quad \text{B/w } X \text{ \& } Y //
 \end{aligned}$$

* Covariance & correlation are used for "linear regression"

* Simple linear regression

correlation & causation

↓
association b/w
& variables
(no dependency)

↓
dependency b/w 2 variables
→ 1 variable result will be
changed based on another variable

* Building simple Linear Regression Model

↳ Regression is a statistical process used to build the models

* Linear regression is used in ML to build models.

① collect data

② preprocessing the data (duplicates removed,
filling null values with mean,
clean the data set)

→ Then dataset will be divided into 2 parts

Training part - 80% (x), Testing part - 20% (y)

⇒ y depends on x.

③ Estimate regression parameters.

$$y = mx + c.$$

$$\boxed{y = \alpha + \beta \cdot x + \epsilon} \text{ (Epsilon)} \Rightarrow \alpha, \beta \text{ are regression parameters.}$$

④ Define the regression function.

⑤ Apply the descriptive statistics.

⑥ validation. (based on Test cases)

* These are the steps to build the regression models

* Simple Linear Regression

* Here we will use only 1 Independent variable
& y is dependent variable

$$y = \alpha + \beta \cdot x + \epsilon$$

$y \rightarrow$ dependent variable / outcome / response variable

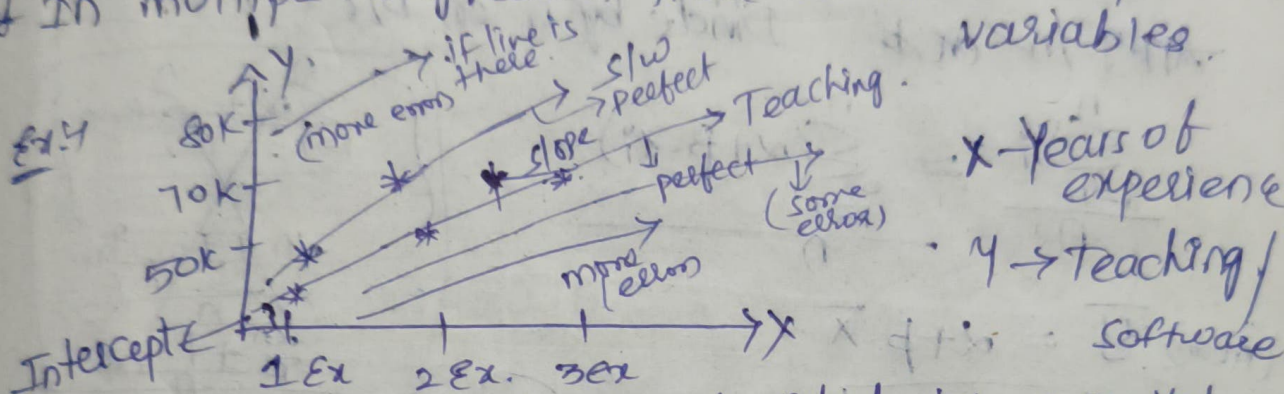
$x \rightarrow$ Independent variable (Explanatory / predictor variable)

$\alpha, \beta \Rightarrow$ regression parameters

$\epsilon \rightarrow$ residual / error (epsilon) $\Rightarrow$ (ideal = 0)

It is used to describe relationship b/w 2 variables (dep & indep)

In multiple regression there are more independent variables.



The line fitting will be decided by α, β values.

$$\begin{aligned} \text{Sum of squares of error} &= \sum_{i=1}^n (\epsilon_i)^2 = \epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \dots + \epsilon_n^2 \\ &= 0 \text{ (for ideal) No error} \\ &\neq 0 \text{ (There is some error)} \end{aligned}$$

* Estimate the parameters by using ordinary least squares (OLS)

$$y_i = \alpha + \beta \cdot x_i + \epsilon$$

$$\epsilon = y_i - \alpha - \beta \cdot x_i \Rightarrow \epsilon = y_i - (\alpha + \beta \cdot x_i) \quad \text{--- (1)}$$

$$\alpha = \text{Intercept}, \beta = \text{slope}$$

$$\min \sum_{i=1}^n (\epsilon_i)^2 = \min \sum_{i=1}^n (y_i - (\alpha + \beta \cdot x_i))^2 \quad \text{--- (2)}$$

Partial derivation w.r. to α .

$$0 = \frac{\partial}{\partial \alpha} \sum_{i=1}^n (y_i - (\alpha + \beta \cdot x_i))^2$$

$$0 = \sum_{i=1}^n 2(y_i - (\alpha + \beta \cdot x_i))(-1)$$

$$\sum_{i=1}^n (-2) \cdot (y_i - (\alpha + \beta x_i)) = 0$$

$$(-2n) \cdot \sum_{i=1}^n (y_i - (\alpha + \beta x_i)) = 0$$

$$\sum_{i=1}^n (y_i - (\alpha + \beta x_i)) = 0$$

$$\sum_{i=1}^n y_i = \sum_{i=1}^n (\alpha + \beta x_i)$$

Apply in α & β Divide by n on b/c

$$\frac{\sum_{i=1}^n \alpha}{n} + \frac{\beta \sum_{i=1}^n x_i}{n}$$

$$= \frac{\alpha \cdot n}{n} + \beta \bar{x}$$

$$\uparrow = \alpha + \beta \bar{x}$$

$$\frac{\sum_{i=1}^n y_i}{n} = \frac{\sum_{i=1}^n (\alpha + \beta x_i)}{n}$$

$$(\text{mean}) \bar{y} = \alpha + \beta \bar{x}$$

$$\boxed{\alpha = \bar{y} - \beta \bar{x}} \quad \text{--- (2)}$$

$$\left(\frac{\sum_{i=1}^n \alpha + \sum_{i=1}^n \beta x_i}{n} \right)$$

$$\frac{\alpha n + \beta n \cdot \sum_{i=1}^n x_i}{n}$$

$$\frac{\alpha n + \beta n \cdot \sum_{i=1}^n x_i}{n}$$

Now, partial derivative (1) w.r. to β

$$0 = \frac{\partial}{\partial \beta} \cdot \sum_{i=1}^n (y_i - (\alpha + \beta x_i))^2$$

$$\sum_{i=1}^n 2(y_i - (\alpha + \beta x_i)) \cdot (-x_i) = 0$$

$$\sum_{i=1}^n (-2)(y_i - (\alpha + \beta x_i))(x_i) = 0$$

$$0 = \frac{\partial}{\partial \beta} \left(\sum_{i=1}^n (y_i - \bar{y} + \beta \bar{x} - \beta x_i)^2 \right) \quad (\text{Replace } \alpha \text{ value})$$

$$\frac{\partial}{\partial \beta} \sum_{i=1}^n (y_i - \bar{y} + \beta(\bar{x} - x_i))^2 = 0$$

$$\frac{\partial}{\partial \beta} \sum_{i=1}^n (y_i - \bar{y} - \beta(x_i - \bar{x}))^2 = 0$$

$$= \sum_{i=1}^n 2(y_i - \bar{y} - \beta(x_i - \bar{x})) \cdot (-(x_i - \bar{x})) = 0$$

$$\sum_{i=1}^n (-2) (y_i - \bar{y} - \beta(x_i - \bar{x}))(x_i - \bar{x}) = 0$$

$$\sum_{i=1}^n [(y_i - \bar{y})(x_i - \bar{x}) - \beta(x_i - \bar{x})^2] = 0$$

$$\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n \beta(x_i - \bar{x})^2$$

$$\beta = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{--- (3)}$$

$$\beta = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) / n}{\sum_{i=1}^n (x_i - \bar{x})^2 / n} = \boxed{\frac{\text{COV}(x, y)}{\text{Var}(x)}}$$

$$\therefore \boxed{\beta = \frac{\text{COV}(x, y)}{\text{Var}(x)}} \quad \text{--- (4)}$$

$$\beta = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) / n}{\sum_{i=1}^n (x_i - \bar{x})^2 / n} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \frac{\sum_{i=1}^n (x_i y_i - \bar{x} y_i - \bar{y} x_i + \bar{x} \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})}$$

$$= \frac{\sum x_i y_i - \bar{x} \sum y_i - \bar{y} \sum x_i + \bar{x} \bar{y} \sum 1}{\sum (x_i - \bar{x})(x_i - \bar{x})}$$

$$= \frac{\sum x_i y_i - \frac{\sum x_i}{n} \sum y_i - \frac{\sum y_i}{n} \sum x_i + n \cdot \bar{y} \cdot \bar{x}}{\sum (x_i^2 - \bar{x} x_i) + \underbrace{\sum ((\bar{x})^2 - \bar{x} x_i)}_0}$$

$$= \frac{\sum x_i y_i - \frac{\sum x_i}{n} \cdot \sum y_i - \frac{\sum y_i}{n} \cdot \sum x_i + n \frac{\sum x_i}{n} \frac{\sum y_i}{n}}{\sum (x_i^2 - \bar{x} x_i) = \underbrace{\sum (\bar{x})^2 - \bar{x} x_i}_0}$$

$$= \frac{\sum x_i y_i - \frac{\sum x_i}{n} \sum y_i}{\sum x_i^2 - \bar{x} \sum x_i}$$

Multiply with n on numerator & denominator.

$$= \frac{n \left(\sum x_i y_i - \frac{\sum x_i}{n} \sum y_i \right)}{n \left(\sum x_i^2 - \bar{x} \sum x_i \right)} \rightarrow \frac{\sum x_i y_i - \bar{x} \sum x_i}{\sum x_i^2 - \bar{x} \sum x_i}$$

$$\beta = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2} \quad \text{--- (5)}$$

$$\therefore y_i = \alpha + \beta \cdot x_i + \epsilon$$

$$\epsilon = y_i - (\alpha + \beta \cdot x_i)$$

$$\alpha = \bar{y} - \beta \cdot \bar{x}$$

$$\beta = \frac{\text{cov}(x, y)}{\text{var}(x)} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

① Find the linear regression eqⁿ for the following data

x	y
2	3
4	7
6	5
8	10

$$y_i = \alpha + \beta \cdot x_i + \epsilon$$

$$[\epsilon = 0]$$

$$y_i = \alpha + \beta \cdot x_i$$

$$\alpha = \bar{y} - \beta \cdot \bar{x}$$

$$\beta = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

X	Y	X.Y	X ²
2	3	6	4
4	7	28	16
6	5	30	36
8	10	80	64
$\sum x_i = 20$	$\sum y_i = 25$	$\sum x_i y_i = 144$	$\sum x_i^2 = 120$

$$\beta = \frac{4(144) - 20 \cdot 25}{4(120) - (20)^2} = \frac{576 - 500}{480 - 400} = \frac{76}{80} = \boxed{0.95} //$$

$$\alpha = \bar{y} - \beta \bar{x}$$

$$6.25 - 0.95(5)$$

$$\boxed{\alpha = 1.5}$$

$$\therefore \boxed{y_i = 1.5 + 0.95 \cdot x_i} \Rightarrow \text{linear regression eqn} //$$

* Main aim of SLR is minimizing the error.

* It is used to predict the next values // continue values //

② find the simple linear regression eqn for the data

X	1	2	3	4	5
Y	50	55	60	65	70

soln

X	Y	X.Y	X ²
1	50	50	25 1
2	55	110	4
3	60	180	9
4	65	260	16
5	70	350	25
$\sum x_i = 15$	$\sum y_i = 300$	$\sum x_i y_i = 950$	$\sum x_i^2 = 55$

$$\beta = \frac{n \sum x_i y_i - \sum x_i \cdot \sum y_i}{n \sum x_i^2 - (\sum x_i)^2} = \frac{5(950) - 15(300)}{5(55) - (15)^2}$$

$$= \frac{4750 - 4500}{275 - 225} = \frac{250}{50} = 5$$

$$\boxed{\beta = 5}$$

$$\alpha = \bar{y} - \beta \bar{x} = 60 - 5(3) = 60 - 15 = \boxed{45}$$

$$\therefore \boxed{y_i = 45 + 5x_i} \Rightarrow \text{linear regression eqn.}$$

$$x=6 \Rightarrow y_i = 45 + 30 = 75 \Rightarrow \boxed{x=6, y=75}$$

* Multiple Linear regression

* SLR tells relⁿship b/w 1 dependent variable & 1 indep variable

* MLR tells relⁿship b/w 1 dep variable & more than 1 indep. variable.

$$\text{Eqn: } \boxed{y_i = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon}$$

dependent / output variable
Independent variables
error/residuals

* $\beta_1, \beta_2, \dots, \beta_n, \alpha \Rightarrow$ regression (parameters / coefficient)

$\alpha \rightarrow$ Intercept

$\beta_1, \beta_2, \dots, \beta_n \rightarrow$ slope.

$$\boxed{\epsilon = y_i - (\alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}$$

$$\boxed{\alpha = \bar{y} - (\beta_1 \bar{x}_1 + \beta_2 \bar{x}_2 + \dots + \beta_n \bar{x}_n)}$$

$$\boxed{\beta_1 = \frac{\sum x_2^2 \cdot \sum x_1 y_1 - \sum (x_1 x_2) \cdot \sum x_2 y}{\sum x_1^2 \cdot \sum x_2^2 - (\sum (x_1 x_2))^2}}$$

$$\boxed{\beta_2 = \frac{\sum x_1^2 \cdot \sum x_2 y - \sum x_1 x_2 \cdot \sum x_1 y}{\sum x_1^2 \cdot \sum x_2^2 - (\sum (x_1 x_2))^2}}$$

* In Machine Learning, the eqⁿ is found by:

x_1	x_2	y
1	4	1
2	5	6
3	8	8
4	2	12

1. Write in Matrix format. (take 4 1's in 1st col

2nd col $\rightarrow x_1$
3rd col $\rightarrow x_2$

$$X = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 6 \\ 8 \\ 12 \end{bmatrix}$$

for α, β_1, β_2

$$\beta = ((X^T \cdot X)^{-1} X^T) \cdot y \rightarrow \text{then we get matrix of } \beta$$

Ex: $\beta = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \rightarrow \begin{matrix} \beta_0 = \alpha \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{matrix}$ } These all substituted in formula of MLR.

Ex: find the MLR eqⁿ for the following data.

x_1	x_2	y
1	4	1
2	5	6
3	8	8
4	2	12

$$\begin{aligned} \alpha &= -1.69 \\ \beta_1 &= 3.48 \\ \beta_2 &= -0.05 \end{aligned}$$

Soln

x_1	x_2	y	x_1^2	x_2^2	$x_1 \cdot y$	$x_2 \cdot y$	$x_1 x_2$
1	4	1	1	16	1	4	4
2	5	6	4	25	12	30	10
3	8	8	9	64	24	64	24
4	2	12	16	4	48	24	8
Σx_1 = 10	Σx_2 = 19	Σy = 27	Σx_1^2 = 30	Σx_2^2 = 109	$\Sigma x_1 y$ = 85	$\Sigma x_2 y$ = 122	$\Sigma x_1 x_2$ = 46

$$\beta_1 = \frac{(109)(85) - 46(122)}{30(109) - (46)^2} = \frac{9265 - 5612}{3270 - 2116} = \frac{3653}{1154} = 3.16$$

$$\beta_2 = \frac{30(122) - 46(85)}{30(109) - (46)^2} = \frac{3660 - 3910}{3270 - 2116}$$

$$= \frac{-250}{1154} = -0.216$$

$$= \boxed{-0.22}$$

$$\alpha = \bar{y} - (\beta_1 \bar{x}_1 + \beta_2 \bar{x}_2)$$

$$= 6.75 - (3.16(2.5) + (-0.22)(4.75))$$

$$= 6.75 - (7.9 - 1.045)$$

$$\alpha = \boxed{-2.195}$$

$$y_i = \alpha + \beta_1 x_1 + \beta_2 x_2$$

$\therefore$ MLR eqⁿ is:

$$y_i = -2.195 + 3.16x_1 - 0.22x_2$$

* Validations of S.L.R

* By using some validations we can validate a model.

- ① co-efficient of determination (R^2)
- ② Hypothesis Test
- ③ Residual Analysis.

④ Outlier Analysis.

① coefficient of determination relationship b/w 1 indep & 1 dep var

* S.L.R $\Rightarrow y = \alpha + \beta \cdot x + E$

* M.L.R $\Rightarrow y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + E$ relationship b/w 1 dep & multiple indep var

* By using these we can validate S.L.R Model.

$$y_i = \alpha + \beta \cdot x_i + E$$

variance in y explained by model

variance in y unexplained by model.

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\bar{y} - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y})^2$$

total variance = sum of squares of total variance

predicted values (explained by model) = mean

sum of sq. of error = sum of squares of regression model

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

s. of sq of
total variance.

s. of sq of
Regression model

sum of sq of error

$\hat{y}$ - predicted value

y_i - actual

$\bar{y}$ - mean

$$SSR = SST - SSE$$

$$R^2 = \frac{\text{variance in } y \text{ is explained by model}}{\text{total variance}}$$

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST}$$

$$R^2 = 1 - \frac{SSE}{SST}$$

Based on R^2 we can validate S.L.R model

$$R^2 \Rightarrow \text{range } 0 \text{ to } 1$$

* High R^2 value $\Rightarrow$ Better line fit (model is fit)

* near to 0 $\Rightarrow$ No line fit (model is not fit)

$$R^2 = r^2 \quad (\text{pearson corr}^n \text{ coefficient})$$

② Hypothesis Test

By using this we can validate the model.

H_0 : There is no diff b/w x & y } S.L. $\Rightarrow$ one sample

H_1 : There is a diff b/w x & y

* we need to use 1 sample t-test.

$$t_s = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$$

$$y = \alpha + \beta \cdot x + \epsilon \Rightarrow y \text{ depends on } x. \quad \epsilon = 0 \Rightarrow y = \alpha + \beta \cdot x$$

If $x=0$, y is no change.

$y_i = \beta_0 + \beta_1 x_i \Rightarrow$ Based on β_1 we need to frame H_0 & H_1 .

$$H_0 = \beta_1 = 0 \quad H_1: \beta_1 \neq 0$$

Based on SLR $\Rightarrow$ $t_s = \frac{\hat{\beta}_1 - \beta_1}{S.E.}$ predicted variable $\rightarrow$ std error

$$t_s = \frac{\hat{\beta}_1}{S.E.}$$

$$S.E. = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-2}}$$

b/c there are 2 parameters (β_0, β_1)

* After finding t_{st} , convert it into prob & compares p value with α .

$P < \alpha \rightarrow$ Do not reject H_0 (No relⁿ b/w x & y)

$P > \alpha \rightarrow$ Reject H_0 (relⁿ b/w x & y)

③ Residual Analysis

* Error = $(y_i - \hat{y})$ $\rightarrow$ y_i - actual value
 $\hat{y}$ - predicted value.
 $\downarrow$
 b/w actual & Predict value.

* Sum of Squares of error should be 0.

* Residual follows the Normal distⁿ. then only error b/m 0.

* Residuals are constant.

[maintain same variance]

* It is c/s "Homoscedasticity".

* If variables are homoscedasticity $\rightarrow$ Better fit model.

If variables are Heteroscedasticity $\rightarrow$ Not a better fit model



④ Outlier Analysis

* 1st we need to identify large variance variables [0-marks, 15-marks students are outliers]

* If outliers are there, the model is not a better fit.

* we need to identify outliers & remove them to make the model better fit.

* Based on these 4 $\rightarrow$ we can validate model.

$R^2 \rightarrow$ near to 1 $\rightarrow$ better fit

near to 0 $\rightarrow$ no better fit

H_0 $\rightarrow$ true $\rightarrow$ no diff b/w X & Y
false $\rightarrow$ diff b/w X & Y

Residual $\rightarrow$ follow normal distn $\rightarrow$ better fit
not follow normal distn $\rightarrow$ no better fit

Outlier $\rightarrow$ there are outliers $\rightarrow$ better fit
more outliers $\rightarrow$ no better fit

* In MLR also similar are used, but instead of t-test, we need to use Anova test b/c more variables.

* Multi collinearity $\rightarrow$ 2 or more variables.

If X_1, X_3 are same $\rightarrow$ collinearity

* There is high correlⁿ b/w X_1 & X_3 .

X_1	X_2	X_3	Y
1	4	1	2
2	5	2	3
3	6	3	4
4	7	4	

* we need to identify those variables & remove them to make model as a better fit.